



Large language models and women's health: a digital companion for informed decision-making

Florian Recker¹ · Ricarda Neubauer¹ · Agnes Wittek¹ · Nadine Scholten²

Received: 27 February 2025 / Accepted: 13 May 2025 / Published online: 21 June 2025
© The Author(s) 2025

Abstract

The integration of artificial intelligence (AI), particularly large language models (LLMs) like ChatGPT, in gynecology and obstetrics has the potential to significantly transform patient care. These AI-driven tools provide continuous access to information on sensitive health topics such as menstruation, contraception, pregnancy, and childbirth, thereby strengthening patient knowledge and autonomy. By offering easily comprehensible medical information at any time, ChatGPT enables patients to better prepare for medical consultations and ask informed questions without hesitation. This fosters a more balanced, partnership-based dialog between patients and healthcare professionals. At the same time, the use of ChatGPT in medicine presents certain limitations and challenges. While it can provide general information, it lacks the ability to assess individual health conditions or support personalized medical decision-making. Additionally, concerns regarding data privacy and the potential for misunderstandings must be carefully considered. As a result, ChatGPT should not replace medical consultations but rather serve as a complementary resource to support patient education and decision-making. Overall, ChatGPT introduces new opportunities for communication in gynecology and obstetrics. Future research and technological advancements may help further refine its capabilities, making it a more tailored and reliable tool for patients while ensuring its safe and responsible integration into healthcare systems.

Introduction

Modern medicine is undergoing a profound transformation driven by digital technologies and artificial intelligence (AI). In recent years, AI's role in healthcare has expanded dramatically—from image recognition in radiology to personalized therapies in oncology. These advancements have the potential to not only improve diagnostic and therapeutic efficiency but also reshape communication between patients and healthcare professionals.

One of the most promising AI applications in this context is ChatGPT, a large language model (LLM) trained on vast amounts of text-based data [1]. ChatGPT is capable of responding to a wide range of medical inquiries in natural

language, making health-related information more accessible to patients [2]. Unlike static online resources, ChatGPT enables individuals to obtain immediate, personalized responses to their questions, enhancing health literacy and helping them make more informed decisions.

The objective of this article is to explore the potential applications, benefits, and challenges of ChatGPT and other LLMs in healthcare, with a specific focus on gynecological and obstetric care.

The role of artificial intelligence in medicine, particularly in gynecology and obstetrics

The role of AI in healthcare has evolved significantly over the past decades. The term “Artificial Intelligence” (AI) encompasses a broad spectrum of technologies, including machine learning and deep learning, which rely on artificial neural networks to analyze complex medical data. Within this field, generative AI and large language models, such as ChatGPT, have gained increasing relevance. ChatGPT was introduced by OpenAI in November 2022 and rapidly

✉ Florian Recker
florianrecker@icloud.com

¹ Department of Obstetrics and Prenatal Medicine, University Hospital Bonn, Venusberg Campus 1, 53127 Bonn, Germany

² Department for Psychosomatic Medicine and Psychotherapy, Center for Health Communication and Health Services Research (CHSR), University Hospital Bonn, Bonn, Germany

became one of the fastest-growing AI applications, surpassing 100 million users worldwide by January 2023 [3].

Initially, AI algorithms were primarily developed for analyzing large datasets and assisting in diagnostics. These systems excel in pattern recognition and, in some cases, even outperform human capabilities. Currently, AI-driven analyses are widely used for predicting disease progression, identifying anomalies in medical imaging, and personalizing treatment plans. In oncology, for example, AI models detect tumor cells in radiological images with high precision, thereby improving diagnostic accuracy and efficiency.

Beyond diagnostics, AI plays a critical role in managing electronic health records, monitoring chronic diseases, and even supporting robotic surgery. Recent advances have opened up new applications in patient care, particularly in communication, education and personalized medicine [4].

In gynecology and obstetrics, AI is particularly valuable in imaging, where it can help diagnose congenital conditions and predict and manage pregnancy-related complications [5–7]. AI applications are also being explored in the interpretation of cardiotocography (CTG) results, with the potential to improve fetal monitoring and maternal care [8].

Large language models and physician–patient communication

Traditional patient education relies heavily on written materials such as brochures, books and websites, which provide valuable information, but are often generic and lack personalization. Given the limited time available for medical consultations, patients frequently turn to online resources to seek additional information, but static content cannot offer the interactive, tailored responses that patients often need (see Fig. 1). LLM such as ChatGPT have the potential to revolutionize patient education by transforming these static materials into interactive dialogs. Patients can use ChatGPT to obtain preliminary information about medical conditions, treatment options, and preventive care before, during, or after their consultations, which can help them better understand their health concerns and prepare informed questions for their physician. Many patients struggle to formulate questions during medical appointments or may hesitate to discuss certain topics due to embarrassment, uncertainty, or cultural barriers. ChatGPT can serve as an intermediary, allowing patients to explore their concerns in a private, judgment-free environment before engaging in a conversation with their physician. Additionally, ChatGPT provides on-demand access to information, offering immediate responses to pressing health questions. The anonymity of AI-based interactions such as those with ChatGPT can be particularly beneficial when addressing sensitive or stigmatized health

issues, such as reproductive health, fertility concerns, or sexually transmitted infections.

While ChatGPT enhances access to information and facilitates patient engagement, it does not replace the role of healthcare professionals. Personalized medical advice requires clinical expertise, diagnostic tools, and a thorough understanding of individual health conditions, all of which ChatGPT cannot provide. However, as a supplementary resource, ChatGPT enables patients to engage more actively in their healthcare, empowering them to take a more informed and proactive role in their medical decisions.

A common challenge in medicine is the complexity of medical terminology and processes. Many patients feel overwhelmed and often do not fully understand the meaning of specific terms or diagnostic procedures. This information asymmetry can be mitigated through the use of AI-based applications, enabling patients to engage in discussions with their physicians and participate in shared decision-making [9]. A practical example includes applications such as ChatGPT or Dedalos, which can translate medical discharge summaries into lay-friendly language. Research by Zaretsky and his team has shown that AI-generated information is often perceived as more comprehensible than traditional physician-written discharge letters, though it is sometimes viewed as incomplete [10]. Large language models (LLMs) have the capability to reprocess specialized medical information in a patient-centered manner or translate it into simpler language [10].

ChatGPT is gradually becoming a key player in physician–patient communication, and its relevance is expected to increase in the future [11]. It serves as a complementary information source, assisting patients in engaging with their healthcare providers on a more equal level and enabling them to ask relevant questions based on prior knowledge. This, in turn, enhances patient autonomy and improves health literacy, fostering a better understanding of one's own health. These factors may contribute to more effective collaboration between patients and physicians.

Given the increasing number of treatment options and the growing emphasis on personalized medicine, patient education plays a crucial role in modern healthcare. The goal of patient education is to empower individuals to make well-informed decisions about their health based on personal preferences. However, limited time in clinical practice, the extensive use of medical jargon, and the emotional burden on patients' present barriers to effective communication.

In addition to its applications for patients, AI is also used by physicians, for instance, in generating educational materials to inform patients, which can then be distributed by healthcare professionals. When appropriately prompted, LLMs can tailor the complexity and readability of the information, producing content that aligns with specific linguistic requirements. For example, information can

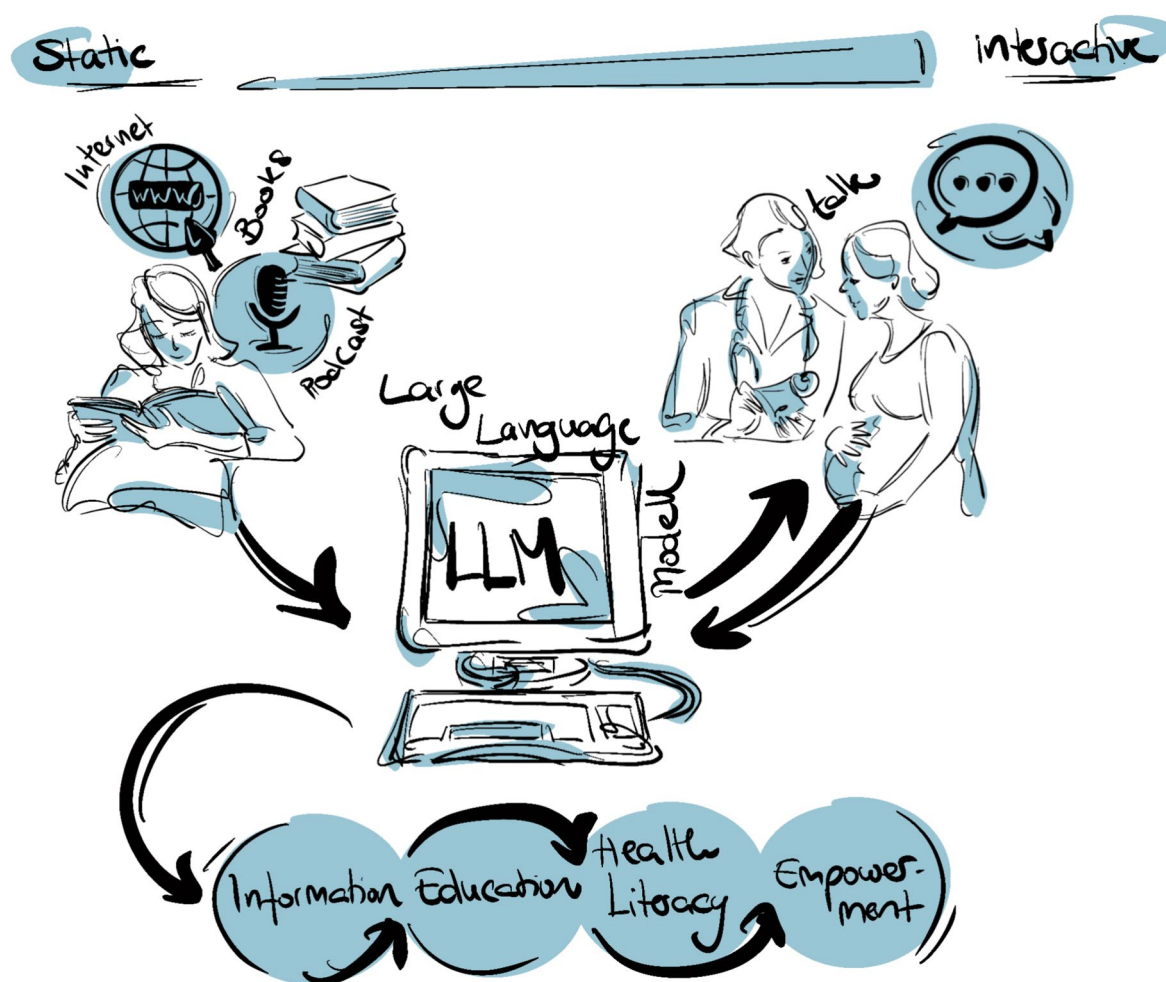


Fig. 1 Potential applications of large language models (LLMs) in gynecology and obstetrics. LLM occupy an intermediate position between traditional unilateral information sources—such as websites, guides, informational brochures, or podcasts—and direct personal interactions with the treating medical team. They bridge the gap between general introductory information and individualized medi-

cal consultation, which considers personal symptoms, limitations, and specific circumstances. By addressing patients' specific concerns, LLMs can enhance individual health literacy, providing a foundation that patients can build upon in subsequent discussions with their healthcare providers

be adapted to a Flesch Reading Ease Score of 60–70, corresponding to an eighth-grade reading level [12].

The potential of large language models in gynecology and obstetrics

In the fields of gynecology and obstetrics, patient education and access to reliable health information are of paramount importance. ChatGPT can play a pivotal role in providing fundamental knowledge on a wide range of topics, including menstruation, contraception, pregnancy, childbirth, and menopause [12]. A recent study by de Vries et al. demonstrated that ChatGPT was able to provide relevant, comprehensible, and largely accurate information on perinatal health topics such as nutrition and warning

signs during pregnancy across multiple languages, highlighting its practical potential in patient education within obstetric care [13].

It can also assist patients in understanding their symptoms by offering explanations for potential causes and recommending subsequent actions. Rotem et al. evaluated ChatGPT's responses to patient questions on urinary incontinence and found that the model achieved favorable ratings in accuracy (71%), comprehensiveness (74%), and safety (74%) among urogynecology experts. Although the authors noted minor inaccuracies and called for further refinement, the results highlight the model's potential as a supportive tool for patient education when used with appropriate oversight and clinical backup [14].

Nevertheless, it is crucial to emphasize that ChatGPT is not intended for medical diagnoses; rather, it is a tool

designed to guide patients in navigating their health concerns.

A key advantage of ChatGPT is its capacity to offer anonymous, round-the-clock support for personal or urgent health inquiries. Many gynecological and obstetric concerns involve complex, deeply personal decisions, such as pregnancy complications or elective termination of pregnancy, where access to neutral, evidence-based information in a private setting can be highly valuable. For instance, some patients may feel uncomfortable discussing abortion-related concerns with their physician due to fear of judgment or social stigma [14]. AI-based resources can help bridge this gap by providing unbiased information that allows patients to make more informed choices.

The quality of AI-generated information depends on its accuracy, clarity, and empathy. Research by Ayers et al. found that AI-generated responses to medical queries were rated significantly higher in empathy compared to responses from human physicians [15]. AI tools like ChatGPT have also demonstrated the ability to simplify complex medical terminology, making health information more comprehensible for patients.

Despite these benefits, the ultimate responsibility for medical accuracy and compliance with legal regulations lies with healthcare professionals. It is the responsibility of healthcare professionals to ensure that AI-generated content aligns with established medical guidelines and does not lead to misinformation [12].

Various LLM models, such as Microsoft Bing, Google Bard, and ChatGPT, have been employed in a Spanish study to generate information on subjects related to gynecology and obstetrics. These include Group B Strep during pregnancy, gestational diabetes, cervical screening, HPV diagnostics, and tetanus-diphtheria-pertussis (Tdap) vaccination during pregnancy. Notably, ChatGPT-generated information was most frequently used by the evaluating gynecologists [16].

Gynecological and obstetric care often involves not only medical but also emotional and psychological challenges, with pregnancy, childbirth, and the postpartum period often accompanied by stress, anxiety, and uncertainty. In this context, ChatGPT can serve as an additional, discreet, and anonymous source of emotional support, helping patients process their emotions and cultivate a positive mindset.

While ChatGPT is not a substitute for psychological counseling, it can help patients express their concerns and develop an initial understanding of their emotional responses. In challenging situations, such as coping with pregnancy loss or receiving a gynecological diagnosis, ChatGPT can serve as an initial point of contact, providing patients with supportive information to help them navigate these difficult experiences.

Examples of potential applications

To illustrate the potential of ChatGPT in gynecology and obstetrics, various examples and case studies can be examined [17]. These cases demonstrate the practical applications of the technology while also highlighting its benefits and challenges:

Example 1: knowledge dissemination and preparation for medical consultations

A pregnant patient preparing for childbirth has numerous questions about different birthing methods and their associated risks, and while her physician provides fundamental information during consultations, some questions remain unanswered. ChatGPT could serve as a supplementary resource, offering additional, easily understandable explanations on natural birth methods, cesarean sections, birthing positions, and pain management options. By exploring the advantages and disadvantages of each method beforehand, the patient can enter subsequent consultations better informed and prepared. In this scenario, ChatGPT enhances the patient's knowledge and health literacy, facilitating more productive communication with healthcare providers.

Example 2: emotional support and empowerment

In the event of a patient encountering a pregnancy complication, they may opt to utilize ChatGPT for information regarding their condition and to assist with preparing for consultations with their physician. Beyond the provision of general details concerning the complication and potential treatment options, ChatGPT is capable of offering emotional support through empathetic responses and reassuring explanations. This can contribute to a reduction in anxiety levels and empower the patient, enabling them to engage in medical discussions with greater confidence and to pose targeted questions. It is important to note that ChatGPT is not a substitute for professional medical consultation and should not be used to make a final diagnosis or determine treatments. These decisions must always be made by a qualified healthcare professional.

AI-powered language models like ChatGPT and similar conversational agents are showing growing potential to emotionally support patients in gynecological and obstetric care. In a retrospective data analysis using *Wysa*, an AI-based self-help chatbot, users who reported

experiencing a maternal event (e.g., pregnancy, birth, or miscarriage) showed a significant reduction in depressive symptoms when they engaged with the chatbot frequently. A qualitative analysis revealed that users turned to the chatbot to express fears, seek emotional relief, and reframe negative thoughts [18]. In a randomized controlled trial by Suharwardy et al., postpartum women who interacted with a cognitive behavioral therapy (CBT) chatbot experienced a greater reduction in depressive symptoms compared to the control group. Additionally, 91% of participants reported being satisfied with the intervention, indicating strong acceptance of the tool [19]. Similarly, during pregnancy, chatbot use has been associated with increased mental well-being: in a qualitative study involving the chatbot *Juno*, pregnant women reported that the intervention helped them pay more attention to their emotional health, although they also wished for more personalization and emotional warmth [20]. Together, these findings suggest that AI-based conversational agents can provide emotional value—by offering reliable information, continuous availability, and a sense of being heard. However, all studies emphasize that such tools should not replace human connection or professional care, but rather be viewed as complementary elements within holistic maternal support systems.

Example 3: assisting in medical decision-making

A young woman seeking information about contraception methods used ChatGPT to explore different options, including birth control pills, intrauterine devices (IUDs), hormone-free alternatives, and potential side effects. ChatGPT may provide comprehensible explanations of the benefits and drawbacks of each method, along with guidance on typical usage recommendations and side effects. This could enable the patient to make a more informed decision and engage in a focused discussion with her physician about her preferences. This example highlights the potential of ChatGPT to enhance patient education and support informed decision-making in gynecology.

Challenges, risks, and limitations of LLMs in medicine

Despite their potential, LLMs must be critically assessed, especially given their inherent limitations. The use of AI in a sensitive domain like gynecology and obstetrics requires careful oversight and regulation to prevent misunderstandings, miscommunication, and potential harm. One key risk associated with ChatGPT is its reliance on publicly available internet data, without the ability to

differentiate between evidence-based information and misinformation. Unlike a physician, ChatGPT does not comprehend medical contexts in a clinical sense; it generates responses based on statistical patterns in its training data. Consequently, AI models may provide general but potentially inaccurate or incomplete information, or even “hallucinate” incorrect statements. Although the study by Rotem et al. found that ChatGPT’s responses were generally well-received, several limitations became evident upon closer analysis. One notable example was the model’s response to a question about the risks associated with urinary incontinence during pregnancy [21]. ChatGPT tends to oversimplify the topic, leaving out important clinical nuances, such as distinguishing between different types of incontinence and the physiological changes that occur during pregnancy, which can affect how symptoms appear and how to manage them. This is particularly concerning when patients lack the medical knowledge to critically assess the accuracy of the information provided. In gynecology and obstetrics, inaccurate or incomplete information can have serious consequences. For instance, if a patient seeks guidance on a rare pregnancy complication or personalized treatment options, ChatGPT may generate insufficient or erroneous responses.

A study by De Vries et al. tested the influence of different languages and phrasing on the results of ChatGPT on perinatal health topics [13]. ChatGPT’s responses varied in accuracy across different languages (Dutch, German, French, and English). While accuracy remained high overall, the study found that non-English responses were sometimes less reliable or less complete, especially in nuanced medical contexts. This reflects a bias in the model’s training data, which is predominantly English, potentially compromising equity in multilingual healthcare access. Furthermore, the study tested how different phrasings (e.g., “Is it safe...” vs. “Is it dangerous...”) influenced the content of ChatGPT’s answers. Even slight variations in how questions were asked led to differences in the responses’ accuracy [13]. In addition, ChatGPT shows a lack of alignment of its recommendations with national care recommendations when communicating its advice to patients. For example, in the Netherlands, pregnant women are often advised to contact a midwife rather than go directly to the hospital. ChatGPT’s default advice conflicted with local guidelines, potentially causing confusion or inappropriate action [13]. Therefore, to ensure safe and effective communication with patients, the use of ChatGPT requires further contextual sensitivity, adaptation to linguistic and cultural differences, and clinical oversight.

Another significant risk is that some patients may misinterpret ChatGPT’s responses as professional medical guidance, despite the fact that it is not a substitute for medical expertise. Patients relying solely on AI-generated responses and foregoing consultation with healthcare professionals could

face serious risks, particularly in cases involving pregnancy complications or medical treatment options. Additionally, data privacy concerns must be addressed when using ChatGPT for health-related inquiries. Patients should be transparently informed about data collection and processing practices to ensure informed consent, which is particularly relevant in gynecology and obstetrics, where many patients value confidentiality regarding their medical information.

To address these challenges, various technical and structural solutions have been proposed. One promising approach involves the use of locally hosted or open-source large language models, which can be deployed within secure healthcare infrastructures to enhance data privacy and minimize the risk of unauthorized data access. These models can be customized to comply with institutional data protection standards, offering a safer alternative to cloud-based AI services. In terms of improving the reliability and medical accuracy of responses, fine-tuning LLMs with domain-specific, peer-reviewed medical data—particularly in the fields of gynecology and obstetrics—can significantly reduce the occurrence of hallucinated or misleading outputs. Additionally, incorporating mechanisms for AI-generated content validation, such as cross-referencing with established clinical guidelines or integrating decision-support layers that flag uncertain or sensitive content, may further increase the trustworthiness of these systems. Alongside technological solutions, it is essential to implement clear user education strategies that emphasize the supportive rather than advisory role of ChatGPT, thereby fostering responsible use and preserving the integrity of the patient-physician relationship.

Ethical considerations

The integration of ChatGPT in medicine gives rise to a number of ethical questions, particularly in relation to patient autonomy and the role of medical professionals. The widespread availability of AI-generated information has the potential to influence patient perceptions of physician expertise, potentially leading to a shift in trust towards healthcare providers and a greater reliance on AI-generated content. A key ethical consideration is to ensure that patient autonomy is genuinely respected, while also recognizing the limitations of ChatGPT as a source of medical advice. Ensuring autonomy necessitates access to a range of reliable information sources, including personalized information collected by ChatGPT and professional medical consultation. ChatGPT should, therefore, be regarded as a supplementary resource, not a replacement for direct medical interactions.

Large language models beyond ChatGPT

While this manuscript focuses primarily on ChatGPT, it is important to consider the broader landscape of large language models (LLMs), which are rapidly evolving and becoming increasingly relevant for healthcare applications. The proprietary models such as Claude (developed by Anthropic) and Gemini (developed by Google DeepMind) provide powerful alternatives to ChatGPT. Claude is particularly notable for its emphasis on constitutional AI and safety orientation, often delivering responses with a cautious and ethical tone—an aspect that can be advantageous when dealing with sensitive medical topics. Gemini, on the other hand, demonstrates strong multimodal capabilities and deep integration with Google's knowledge ecosystem, which could improve access to verified health information across different media formats.

In contrast, open-source models such as Mistral and LLaMA (developed by Meta) offer significant advantages in terms of transparency, customization and local deployment. These models can be fine-tuned to institution-specific or domain-restricted medical datasets, making them particularly attractive for privacy-sensitive environments such as gynecology and obstetrics. For example, Mistral models are known for their efficient architecture (e.g., sparse mixture-of-experts in Mixtral) and competitive performance despite smaller parameter sizes, while LLaMA models are widely used in research due to their high adaptability and open licensing for academic and non-commercial clinical applications. Each of these LLMs differs in key characteristics such as training data transparency, model architecture, reasoning power and alignment strategies—factors that directly influence their suitability for clinical use. Including a broader range of models in future comparative research could provide valuable insights into how different LLMs perform in real-world healthcare scenarios, particularly in terms of accuracy, explainability, and ethical alignment. Such analyses could ultimately inform more nuanced decisions about the safe and effective integration of AI tools into gynecological and obstetric care.

Future perspectives and research potential

The application of ChatGPT in gynecology and obstetrics is still in its early stages, but shows great potential for future development. To maximize its benefits and address current challenges, ongoing research and technological advancements are essential.

A key priority in the future development of ChatGPT is improving its medical accuracy and reliability. Ensuring that responses are based on up-to-date, evidence-based

medical sources is critical. Integration of curated medical databases updated regularly is a potential avenue for future development, providing patients with more reliable and current information. Training AI models with a specific focus on gynecology and obstetrics could further enhance their ability to provide specialized and contextually appropriate responses, but it is essential to clearly communicate the limitations of such models and continuously emphasize the necessity of professional medical consultation.

Cultural and personal factors also play a significant role in medical decision-making within gynecology and obstetrics. Future AI models could be trained to better accommodate individual needs, offering more personalized support. For example, ChatGPT could be developed to consider cultural preferences and religious beliefs, ensuring that responses align with patients' values and circumstances. Another promising direction involves the integration of ChatGPT into existing medical platforms and patient portals. Gynecological clinics, birthing centers, and midwifery practices stand to benefit from incorporating AI-powered chat tools within their digital platforms, providing patients with centralized access to information while facilitating seamless transitions to professional medical consultation when needed.

Conclusion

The incorporation of ChatGPT into gynecology and obstetrics presents a significant opportunity to enhance patient education, improve access to information, and strengthen health literacy. By offering an anonymous, accessible, and comprehensible source of information, it empowers patients to engage more actively in their care and communicate more effectively with healthcare providers. However, it is essential to emphasize that ChatGPT should complement—not replace—medical consultation. Addressing concerns such as data security, medical accuracy, and the risk of misinterpretation through continuous oversight and development is essential. LLMs serve as a conduit between general health information and personalized care, helping patients better understand their symptoms and medical options. By tailoring responses to individual concerns, these models can support informed decision-making and improve communication between patients and healthcare professionals. Future research should prioritize the refinement of AI-based tools to enhance reliability, minimize bias, and ensure ethical and responsible implementation in clinical practice. In the field of gynecology and obstetrics, ChatGPT holds significant potential to transform patient care by fostering shared decision-making and integrating patients as informed partners in treatment. While AI-driven technologies show

promise for improving communication and autonomy in medicine, their long-term impact will depend on further research and responsible integration into healthcare systems.

Author contributions F.R. conceptualized the study, supervised the research, and contributed to manuscript writing and revisions. R.N. contributed to the literature review and drafted sections of the manuscript. A.W. assisted in data interpretation, writing, and provided critical revisions. N.S. contributed to the theoretical framework, reviewed ethical considerations, writing and provided expertise on health communication aspects. All authors read and approved the final manuscript.

Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

References

1. Deng J, Lin Y (2023) The benefits and challenges of ChatGPT: an overview. *FCIS* 2(2):81–83
2. Alowais SA, Alghamdi SS, Alsuehaby N, Alqahtani T, Alshaya AI, Almohareb SN et al (2023) Revolutionizing healthcare: the role of artificial intelligence in clinical practice. *BMC Med Educ* 23(1):689
3. Eysenbach G (2023) The role of ChatGPT, generative language models, and artificial intelligence in medical education: a conversation with ChatGPT and a call for papers. *JMIR Med Educ* 6(9):e46885
4. Dave T, Athaluri SA, Singh S (2023) ChatGPT in medicine: an overview of its applications, advantages, limitations, future prospects, and ethical considerations. *Front Artif Intell* 4(6):1169595
5. Kim HY, Cho GJ, Kwon HS (2023) Applications of artificial intelligence in obstetrics. *Ultrasonography* 42(1):2–9
6. Malani SN, Shrivastava D, Raka MS (2023) A comprehensive review of the role of artificial intelligence in obstetrics and gynecology. *Cureus*. Available from: <https://www.cureus.com/articles/122226-a-comprehensive-review-of-the-role-of-artificial-intelligence-in-obstetrics-and-gynecology> Feb 12 [cited 2025 Feb 13]
7. Sarno L, Neola D, Carbone L, Saccone G, Carlea A, Miceli M et al (2023) Use of artificial intelligence in obstetrics: not quite ready for prime time. *Am J Obstet Gynecol MFM* 5(2):100792

8. Aeberhard JL, Radan AP, Delgado-Gonzalo R, Strahm KM, Sigurthorsdottir HB, Schneider S et al (2023) Artificial intelligence and machine learning in cardiotocography: ascoping review. *Eur J Obstet Gynecol Reprod Biol* 281:54–62
9. Mai S, Chang L, Xu RH, Su S, Wang D (2024) Doctor interaction behavior, patient participation in value co-creation and patient satisfaction: cross-sectional survey in a tertiary-level hospital from Guangzhou, China. *Sci Rep* 14(1):23025
10. Zaretsky J, Kim JM, Baskharoun S, Zhao Y, Austrian J, Aphinyanaphongs Y et al (2024) Generative artificial intelligence to transform inpatient discharge summaries to patient-friendly language and format. *JAMA Netw Open* 7(3):e240357
11. Clusmann J, Kolbinger FR, Muti HS, Carrero ZI, Eckardt JN, Laleh NG et al (2023) The future landscape of large language models in medicine. *Commun Med* 3(1):141
12. Grünebaum A, Chervenak J, Pollet SL, Katz A, Chervenak FA (2023) The exciting potential for ChatGPT in obstetrics and gynecology. *Am J Obstet Gynecol* 228(6):696–705
13. De Vries PLM, Baud D, Baggio S, Ceulemans M, Favre G, Gerbier E et al (2025) Enhancing perinatal health patient information through ChatGPT—An accuracy study. *PEC Innovation* 6:100381
14. Chor J, Tusken M, Lyman P, Gilliam M (2016) Factors shaping women's pre-abortion communication with their regular gynecologic care providers. *Women's Health Issues* 26(4):437–441
15. Ayers JW, Poliak A, Dredze M, Leas EC, Zhu Z, Kelley JB et al (2023) Comparing physician and artificial intelligence Chatbot responses to patient questions posted to a public social media forum. *JAMA Int Med* 183(6):589
16. Solmonovich RL, Kouba I, Quezada O, Rodriguez-Ayala G, Rojas V, Bonilla K et al (2024) Artificial intelligence generates proficient Spanish obstetrics and gynecology counseling templates. *AJOG Global Rep* 4(4):100400
17. Matsubara S, Matsubara D (2024) ChatGPT use in daily obstetrics and gynecology practice: clarification and recommendations. *Arch Gynecol Obstet* 310(4):2295–2296
18. Inkster B, Kadaba M, Subramanian V (2023) Understanding the impact of an AI-enabled conversational agent mobile app on users' mental health and wellbeing with a self-reported maternal event: a mixed method real-world data mHealth study. *Front Glob Womens Health* 2(4):1084302
19. Suharwardy S, Ramachandran M, Leonard SA, Gunaseelan A, Lyell DJ, Darcy A et al (2023) Feasibility and impact of a mental health chatbot on postpartum mental health: a randomized controlled trial. *AJOG Glob Rep* 3(3):100165
20. Mancinelli E, Magnolini S, Gabrielli S, Salcuni S (2024) A Chatbot (Juno) prototype to deploy a behavioral activation intervention to pregnant women: qualitative evaluation using a multiple case study. *JMIR Form Res* 14(8):e58653
21. Rotem R, Zamstein O, Rottenstreich M, O'Sullivan OE, O'reilly BA, Weintraub AY (2024) The future of patient education: A study on AI -driven responses to urinary incontinence inquiries. *Intl J Gynecol Obstet* 167(3):1004–1009

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.